User Evaluation (Quantitative)

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Participants

Ethical treatment of participants

- Helsinki Declaration on Ethical Principles for Medical Research Involving Human Subjects

- American Psychological Association's Ethical Principles of Psychologists and Code of Conduct (http://www.apa.org/ethics)

- Code of conduct for the Association for Computing Machinery (https://www.acm.org/code-of-ethics)

Participants - Ethics

Treat participants with respect; value their time, honor their opinions, and take any criticism seriously.

Do not expose participants to dangerous or potentially harmful situations; this includes physical, mental, and emotional concerns.

Make sure participants want to participate; get informed consent (there are templates available online and your institution might also have one).

Participants - Ethics

Make sure that any reimbursement to participants is adequate. Reimbursement should not be necessary for participants to enroll in your study, but it should still compensate participants for their time and any expenses incurred by participating (e.g., transportation costs).

Debrief the participants; explain the purpose of the experiment and answer any questions they might have about your study

Variables

Collecting sensible data: bio/health or personal data need approval from authorities or ethical committees.

Bio/body interventions also need special approval from authorities or ethical committees.

Participants

Collect and report participants' profile and backgrounds important for your study, for example:

- Age (collect raw data, not intervals)
- Gender
- Education
- Field of work/study (worker vs student)
- Nationality/Cultural background

….

But keep your study **anonymous**. (e.g., P1, P2, P3, etc.)

Task(s)

Tasks can be decided in many ways but shall consider:

- Be representative of what users would do outside the experiment.
- Capture the essence of what is being studied.
- Reduce variation and remove non-essential features.

Dependent Variables

What to measure?

Quantitative data – example:

task completion times,

accuracy or error rates,

number of clicks or steps,

questionnaire answers

Qualitative data – example:

interviews (not so easy formal analysis, and not our focus)

observations (video analysis can also be quantitative)

Quantitative data

Survey/questionnaires: Collect a large sample of structured self-reported data.

Log files: record data about what users do with interactive systems.

(Log files can also be replaced by video recording and analysis, but more time-consuming and less precise – but a good backup plan)

Functional prototypes facilitate the collection of log data

Survey/questionnaires

A questionnaire consists of a series of questions presented and answered in a structured way (different from interviews).

Questionnaires often use Likert scale answers:

"I enjoy reading textbooks in an e-book format on a digital device."

1—Strongly disagree

2—Somewhat disagree

3—Neutral

4—Somewhat agree

5—Strongly agree

Likert scales usually have 5, 7, or 11 points.

Survey/questionnaires

Questionnaires can be used to collect general information but also to evaluate a system, depending on how the questions are made.

You can have your own questions or use standard questionnaires.

Standard questionnaires were already validated and facilitate comparisons to other systems.

Examples: System Usability Scale (SUS), NASA Task Load Index (NASA-TLX), Self-Assessment Manikin (SAM), User Engagement Scale (UES)

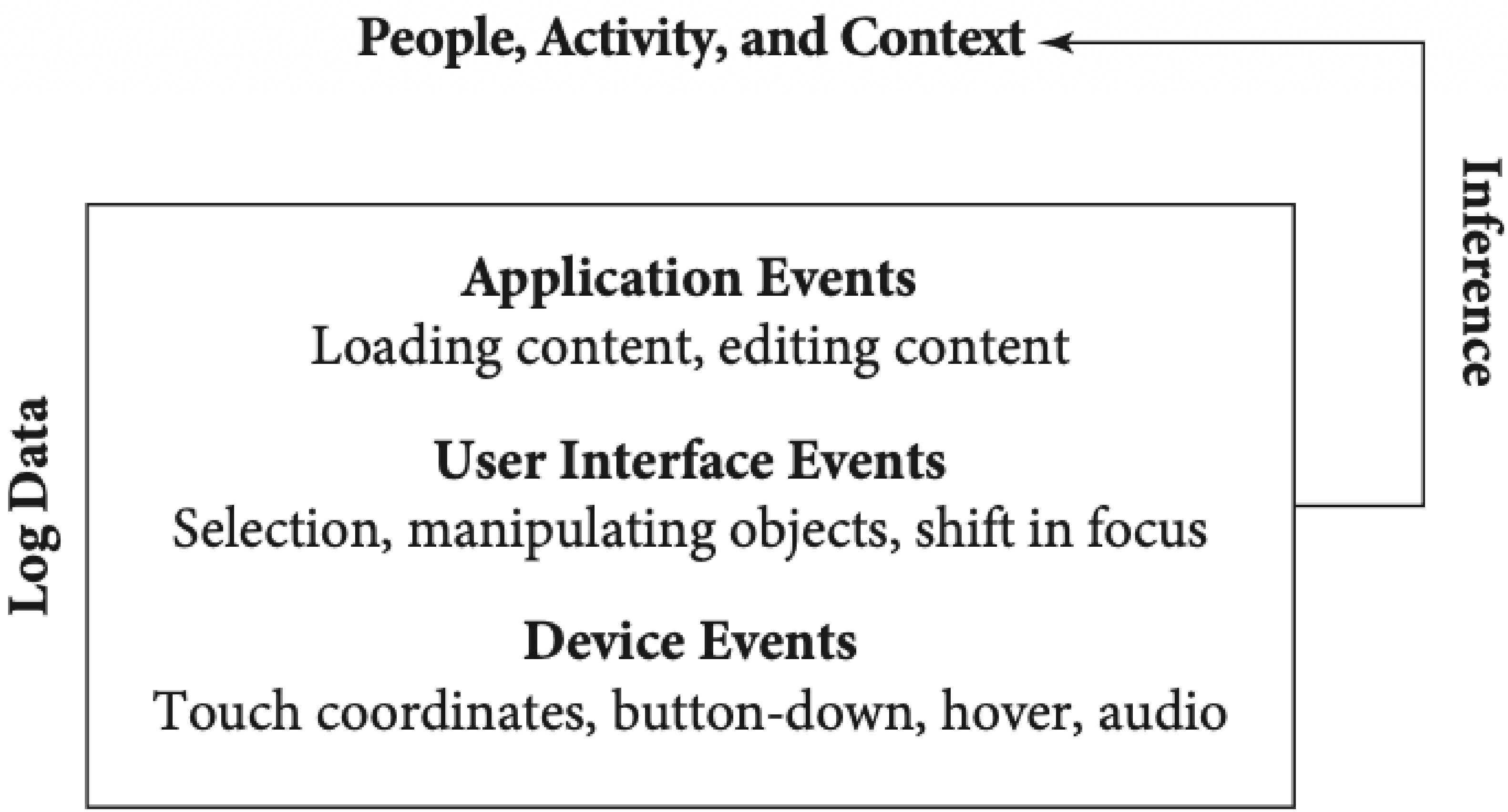
https://www.taylorfrancis.com/chapters/edit/10.1201/9781498710411-35/sus-quick-dirty-usability-scale-john-brooke

Log files

Log file analysis relies on the automatic capture of interactions with interactive systems in a computer file—a log. Logs can be obtained from a variety of systems, including program logs, logs of interactions with web pages, logs of interactions in applications, and so on.

Is a form of noninterventional user research - *unobtrusive research*.

Log Files


People, Activity, and Context
Inference
Log Data
Application Events
Loading content, editing content
User Interface Events
Selection, manipulating objects, shift in focus
Device Events
Touch coordinates, button-down, hover, audio

Results and Analysis - DataTypes

Four levels, or scales, of measurement:

- nominal: a number of distinct classes or categories.
- **ordinal:** the ordinal type allows for rank order, e.g., n. of clicks
- **interval:** length of the scale equal differences between measurements correspond to equal differences in the amount of the attribute being measured. It does not have a true zero point. E.g., Temperatures in °C.
- **ratio:** same as interval but with a true zero point, e.g., weight, height, blood glucose level, as well as measures of certain behavior

Descriptive statistics

Descriptive statistics: describe relationships between variables in the dataset.

Mean/Average (M) = $\frac{1}{n}\sum_{i=1}^{n} a_i = \frac{a_1 + a_2 + \cdots + a_n}{n}$

Median (Mdn) $= \begin{cases} X[\frac{n+1}{2}] & \text{if n is odd} \\ \frac{X[\frac{n}{2}] + X[\frac{n}{2}+1]}{2} & \text{if n is even} \end{cases}$

X = ordered list of values in data set

n = number of values in data set

Standard Deviation (SD)= $\sqrt{\frac{\sum_{i=1}^{n}(x_i - \overline{x})^2}{n-1}}$

x_i = Value of the i^{th} point in the data set

$\overline{x}$ = The mean value of the data set

n = The number of data points in the data set

Mean vs Median

The median is the number that lies in the middle of an ordered dataset that goes from lowest to highest. It shouldn't be confused with the mean that's determined by adding the numbers in a set together and dividing by the total number of data points.

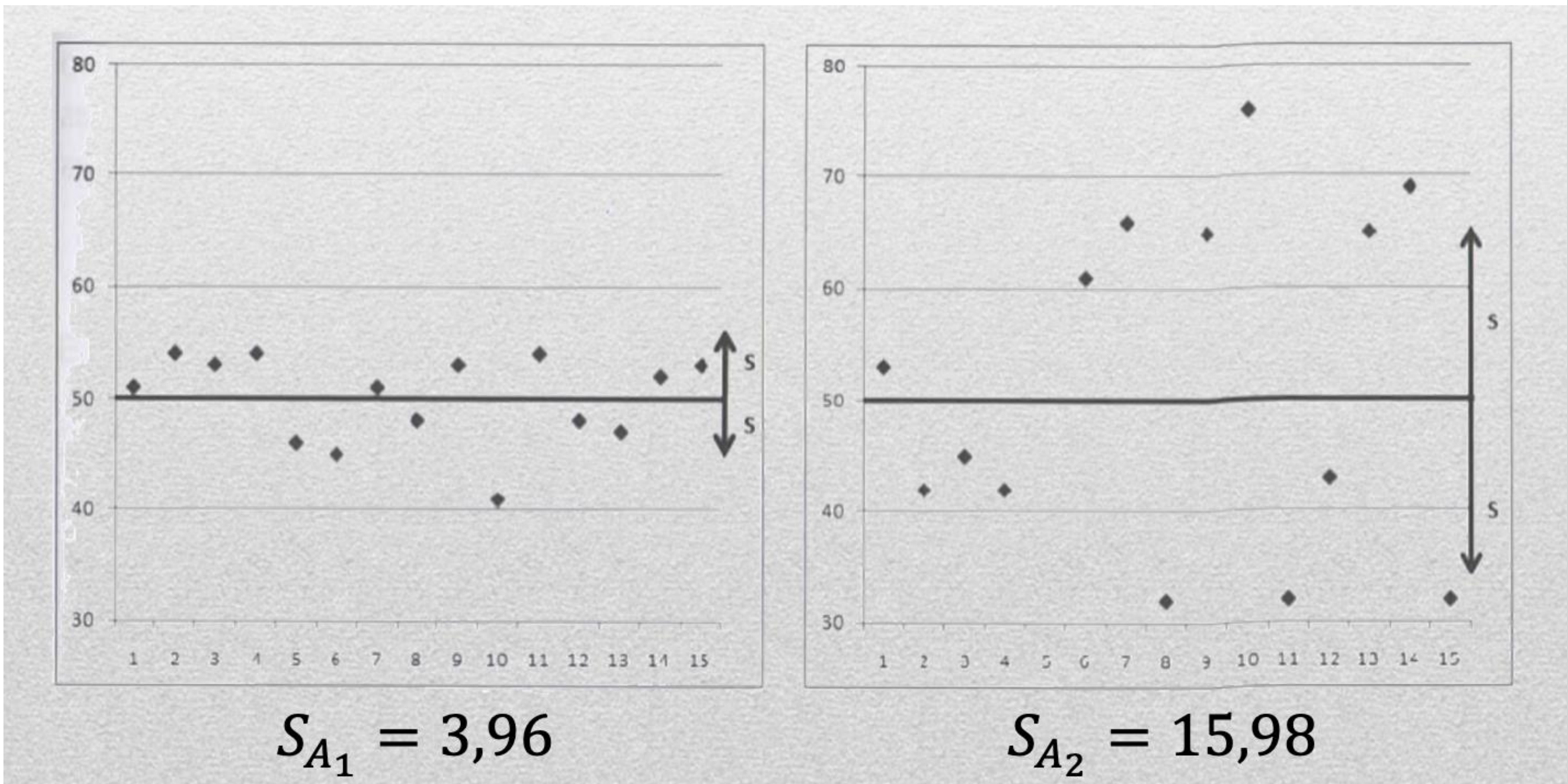
Example: 1, 2, 2, **3**, 4, 7, 9

Mean (M) = 1 + 2 + 2 + 3 + 4 + 7 + 9) / 7 = 4

Median (Mdn) = 3

Standard Deviation

Standard deviation describes how much variability —or fluctuation—exists within a data set.



Standard Deviation vs Standard Error

The standard error indicates how accurately a data set represents the true population by comparing the dataset's average to the population's average.

The size of the data set—the sample size—doesn't affect standard deviation, but the sample size is a key factor in calculating the standard error.

Descriptive statistics – Data types

What to compute and report (APA Style)?

Ordinal: always report Medians (Mdn)

(on the number of clicks there is no meaning for values like 3.2764, interval sizes are subjective)

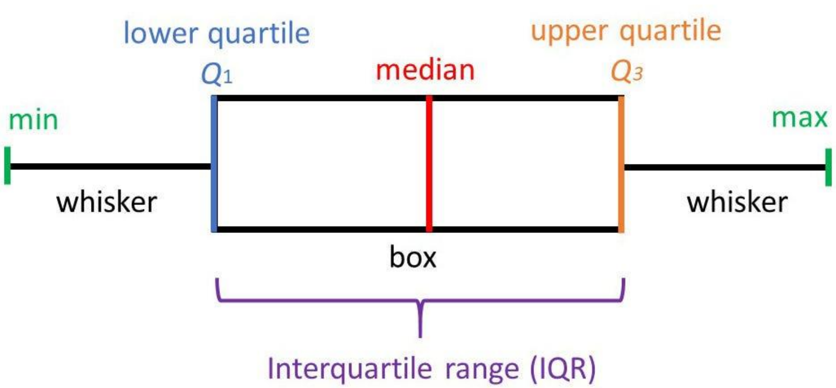
Interval/Ratio: always report Mean (M) + Deviation (SD)

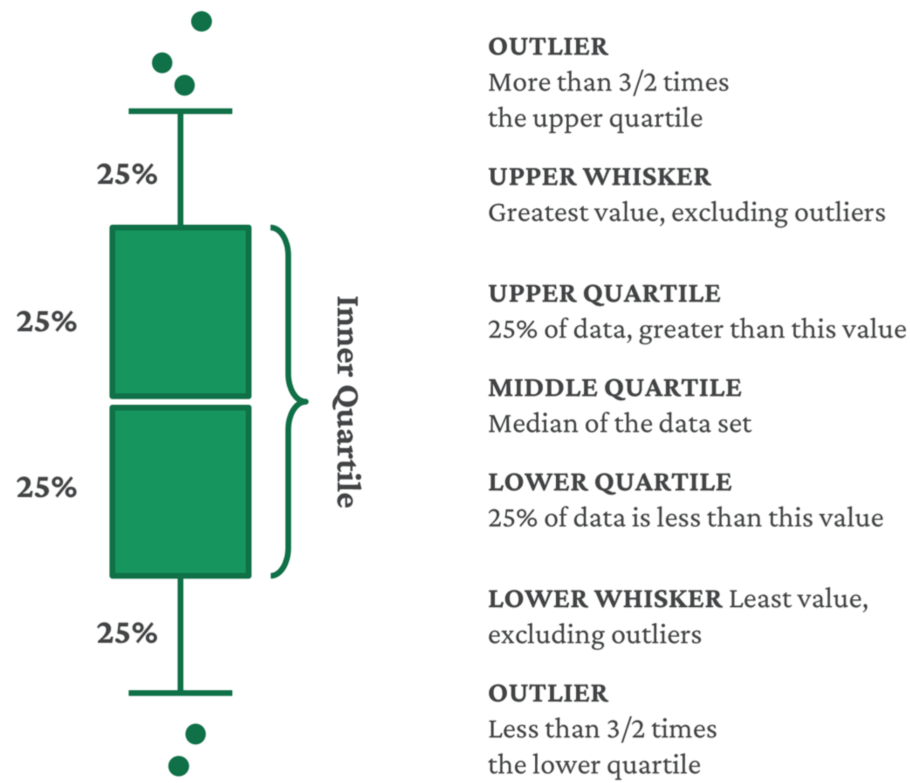
(in some cases, you may also want to report median values)

In both report the size of sample of participants; **N = XX**

Descriptive statistics – Charts

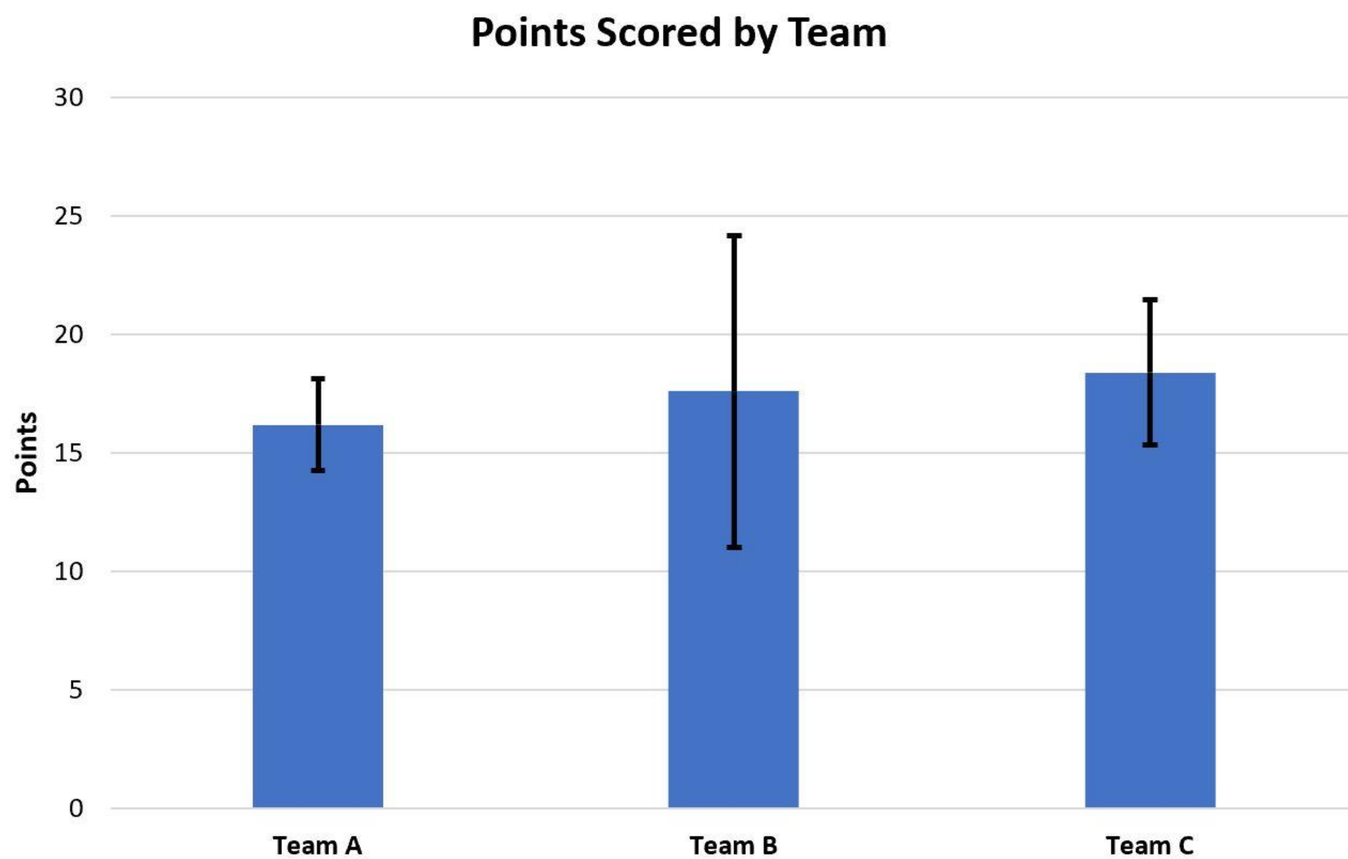
Medians – Boxplots - displays a dataset based on the five-number summary: the minimum, the maximum, the sample median, and the first (lower) and third (upper) quartiles.





Descriptive statistics – Charts

Mean (bar chart) + SD (use error bars to represent it)



Evaluation – Lab vs Field

Laboratory evaluation:

- The responsible for the study can control what happens during the evaluation: the tasks users carry out, the instructions they receive, and other factors that may affect user performance and behavior.
- Responsible for the study can also ensure that observations are independent, that is, that participants do not influence each other.
- These practices minimize the risk of irrelevant factors compromising the validity of the conclusions drawn from a study.

Evaluation – Lab vs Field

Field evaluation:

- Bring out prototypes of interactive systems in their assumed context of use.
- They tend to provide high ecological validity (sometimes also called external validity)
- The conditions in which the study is arranged closely resemble the real-world conditions of use of a system.

Evaluation – Lab vs Field

Field experiments need to balance the causal logic of experiments with the realism of the field.

Experiments in the field, three core assumptions of experimental research are threatened: **random assignment, control, and the independence of observations.**

Evaluation – Lab vs Field

Random assignment: not be able to randomly assign users to particular conditions in the field. Work with users who happen to use a particular user interface or need to carry out particular tasks.

Control: cannot control events to the same extent as in a laboratory setting. Field studies are vulnerable to a surprisingly large number of unexpected events and factors that may substantially affect the outcomes.

Independence of observations: observations may not be independent of each other. For example, if you want to evaluate a system with a school class, the students can—or rather, will—talk about the experiment with each other, which will influence their behavior.

Evaluation – Lab vs Field

Type	Advantage	Explanation
Field		
	Realism	Brings evaluation to the context and activities in which the interactive system will be used
	Sociotechnical fit	Enables the discovery of the fit, or lack thereof, between social activities and structures and the system
	Flexible duration	Field evaluations typically last from a few days to a few months
Lab		
	Precision	It is easy to obtain precise measurements of system use and control for factors that are not studied
	Low cost	Requires fewer resources than a field study
	Can evaluate unusual tasks	Tasks that rarely occur “in the wild” can be arranged
	Does not require a robust system	Studies can be run with rough prototypes

Creating Logs

Record all data in a local text file – Best solution, but be aware of permissions on users' devices (you can use a single device)

Send data to a server for storage – Avoids (some) device permissions, but requires an Internet connection (stable connection or data may be lost). You may need to use a specific data format when sending to the server. Important to rely on device data (not on server data - communication delays).

Prototypes need to be (minimal) robust; if they crash, data will be lost.

Creating Logs

What data to collect?

- There is no rule; it depends on what you want to look
  e.g., if missing targets:
    x,y coordinates, every time a button is pressed

- However, timestamps are always useful
  e.g., task completion times

Creating Logs

In what format?

You choose, but Comma-Separated Values (CSV) files can be easily imported into statistical tools (Excel, SPSS, R, etc.).

Example:

```
timestamp,user_id,action,module
2026-04-21 10:01:05,42,login,auth
2026-04-21 10:02:15,107,view_page,dashboard
2026-04-21 10:05:00,42,update_profile,settings
```

Creating Logs

Writing Text Files

Flutter

https://docs.flutter.dev/cookbook/persistence/reading-writing-files

QT

https://doc.qt.io/qt-6/qfile.html

Unity

https://support.unity.com/hc/en-us/articles/115000341143-How-do-I-read-and-write-data-from-a-text-file

Creating Logs

Be careful about permissions:

https://dineshphalwadiya1995.medium.com/manage-external-storage-using-flutter-getx-93a0ceb917b4